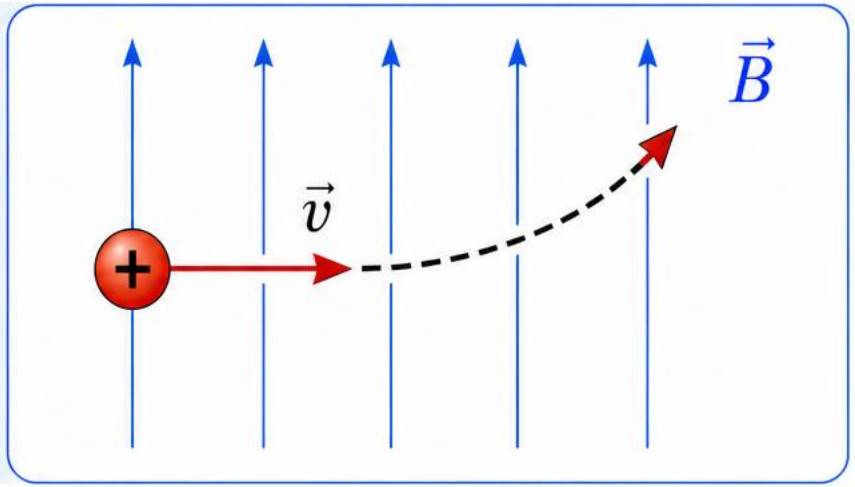
Studying magnetism

1 What are the effects of a magnetic field?

What happens to a charged particle when it is placed in an external magnetic field.



2 What are the sources of a magnetic field?

What creates magnetic fields?

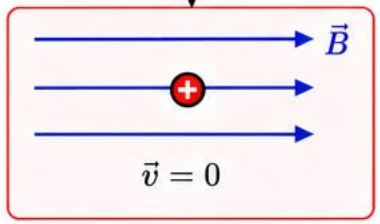
What are the physical origins of $\vec{B}$?

Summary

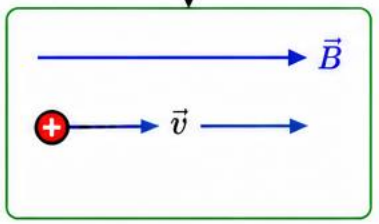
What happens to a charge in a magnetic field?

Is the charge moving?

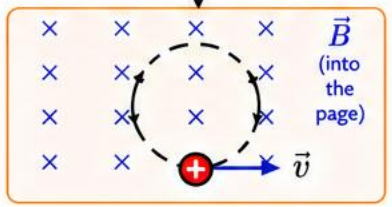
- **NO** → Not moving ($v = 0$)
  - **Result:** ($v = 0 \rightarrow F_B = 0$)
    Nothing happens. The charge stays at rest.



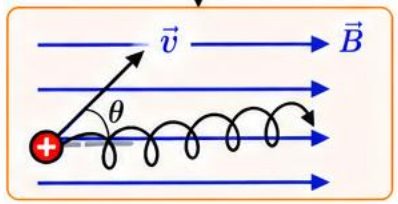
- **YES** → Is the velocity aligned with the magnetic field? (i.e., moving along $\vec{B}$)
  - **YES** → **Result:** ($\vec{v} \neq 0$, $\theta = 0$ or π) ($\sin 0$ or $\sin \pi = 0 \rightarrow F_B = 0$)
    No magnetic force. The charge continues in a straight line.



  - **NO** → Is the velocity perpendicular to the magnetic field?
    - **YES** (perpendicular) → **Result:** Magnetic force acts. The charge moves in a circular path.

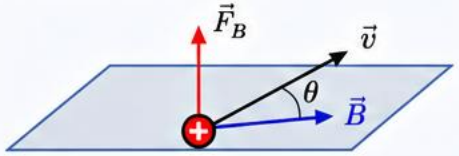


    - **NO** (at an angle) → **Result:** Magnetic force acts. The charge moves in a helical (spiral) path.



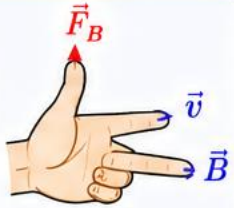
Key idea:

The magnetic force is always perpendicular to the plane formed by $\vec{B}$ and $\vec{v}$.



$$\vec{F}_B = q\vec{v} \times \vec{B}$$
$$= qvB\sin\theta$$

(Direction given by the right-hand rule)



(Index: $\vec{v}$, Middle: $\vec{B}$, Thumb: $\vec{F}_B$)

The magnetic force is like a steering wheel:

The magnetic force doesn't change the speed of the charge, only its direction—just like a steering wheel.

Speed stays the same. Only direction changes.

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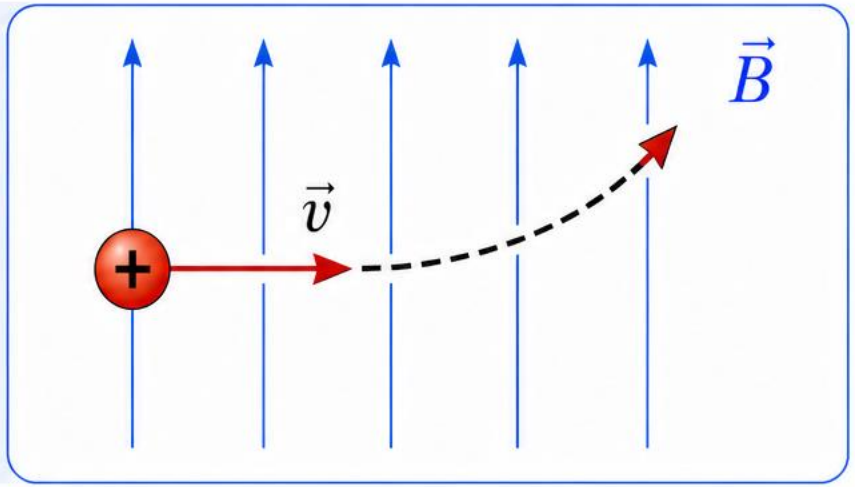

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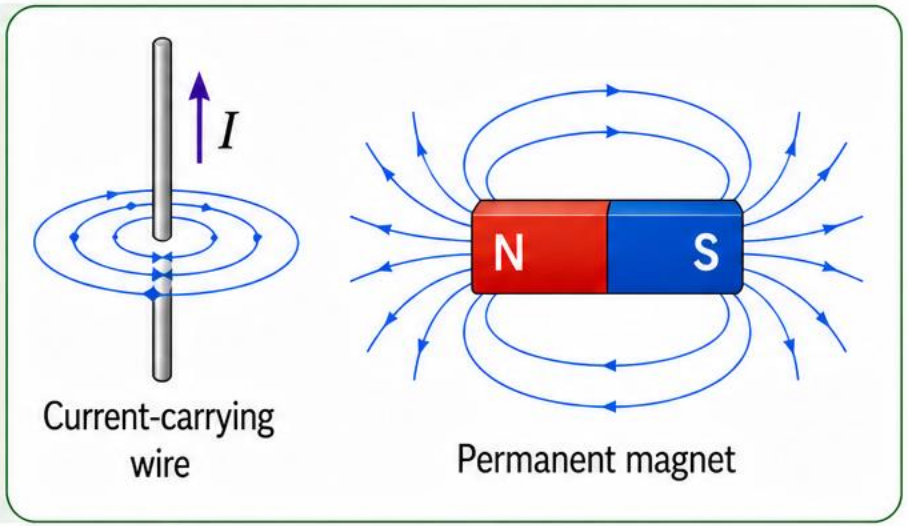
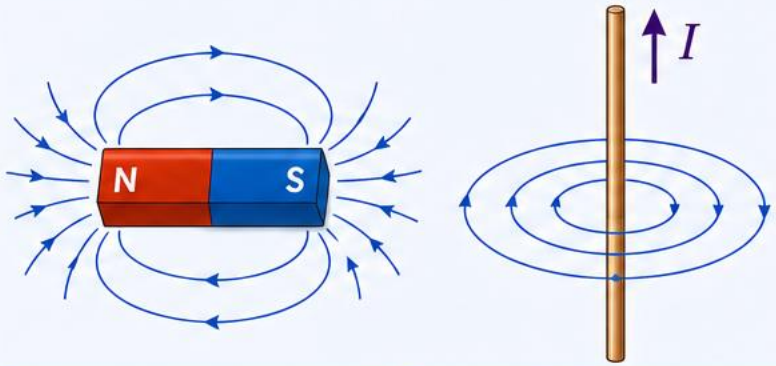


SOURCES OF THE MAGNETIC FIELDS

Magnetic fields are created in two fundamental ways:

1. CURRENTS
(Moving Charges)

Examples: magnets, current-carrying wires.



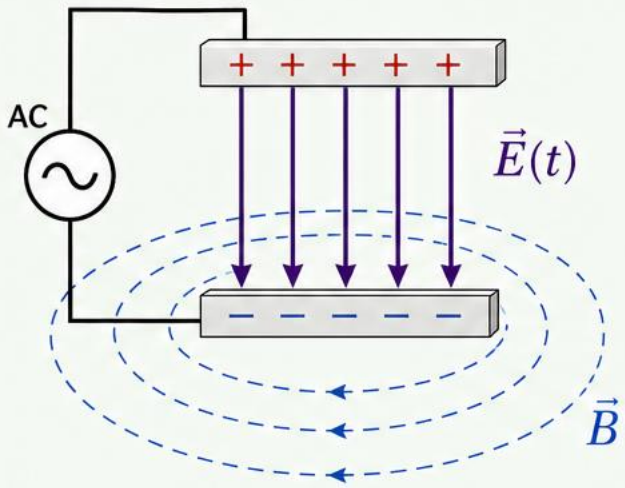
HOW DO WE CALCULATE MAGNETIC FIELDS DUE TO CURRENTS?

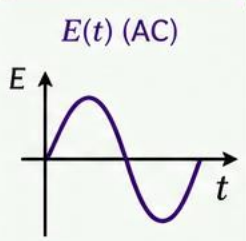
BIOT–SAVART LAW

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{I\, d\vec{\ell} \times \hat{r}}{r^2}$$

2. CHANGING ELECTRIC FIELDS
(Time-Varying Electric Fields)

A changing electric field (with time) also produces a magnetic field.





Electric field varies with time.

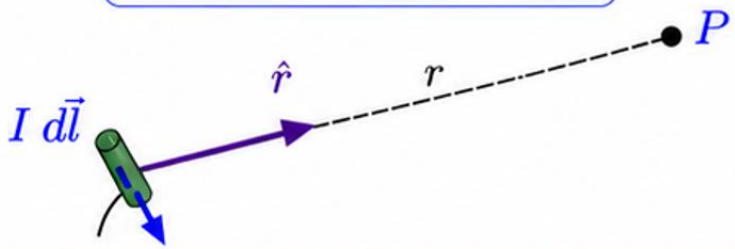
Key idea: Magnetic fields can come from currents (moving charges) or from changing electric fields.
In this chapter, we focus on magnetic fields due to currents which is calculated using Biot–Savart law.

BIOT–SAVART LAW

Differential form of Biot–Savart law gives the magnetic field contribution at a point P due to a tiny current element in a current–carrying wire.

BIOT–SAVART LAW (Differential Form)

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I\, d\vec{l} \times \hat{r}}{r^2}$$



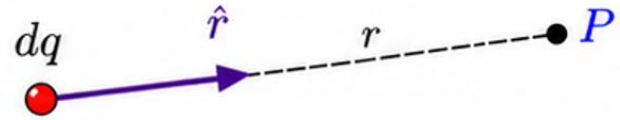
μ_0	=	permeability of free space ($4\pi \times 10^{-7}$ T·m/A)
$I\, d\vec{l}$	=	current element, direction is the direction of the current
$\hat{r}$	=	unit vector from current element to field point P
r	=	distance from current element to field point P

COMPARING BIOT–SAVART AND COULOMB'S LAW

Quantity	Magnetic Field $\vec{B}$	Electric Field $\vec{E}$
Source	Current element $I\, d\vec{l}$	Charge element dq
Direction	Perpendicular to both $I\, d\vec{l}$ and $\hat{r}$ ($I\, d\vec{l} \times \hat{r}$)	Along $\hat{r}$
Dependence on r	$\propto \frac{1}{r^2}$	$\propto \frac{1}{r^2}$
Constant	$\frac{\mu_0}{4\pi}$	$\frac{1}{4\pi\epsilon_0}$

COULOMB'S LAW (For Reference)

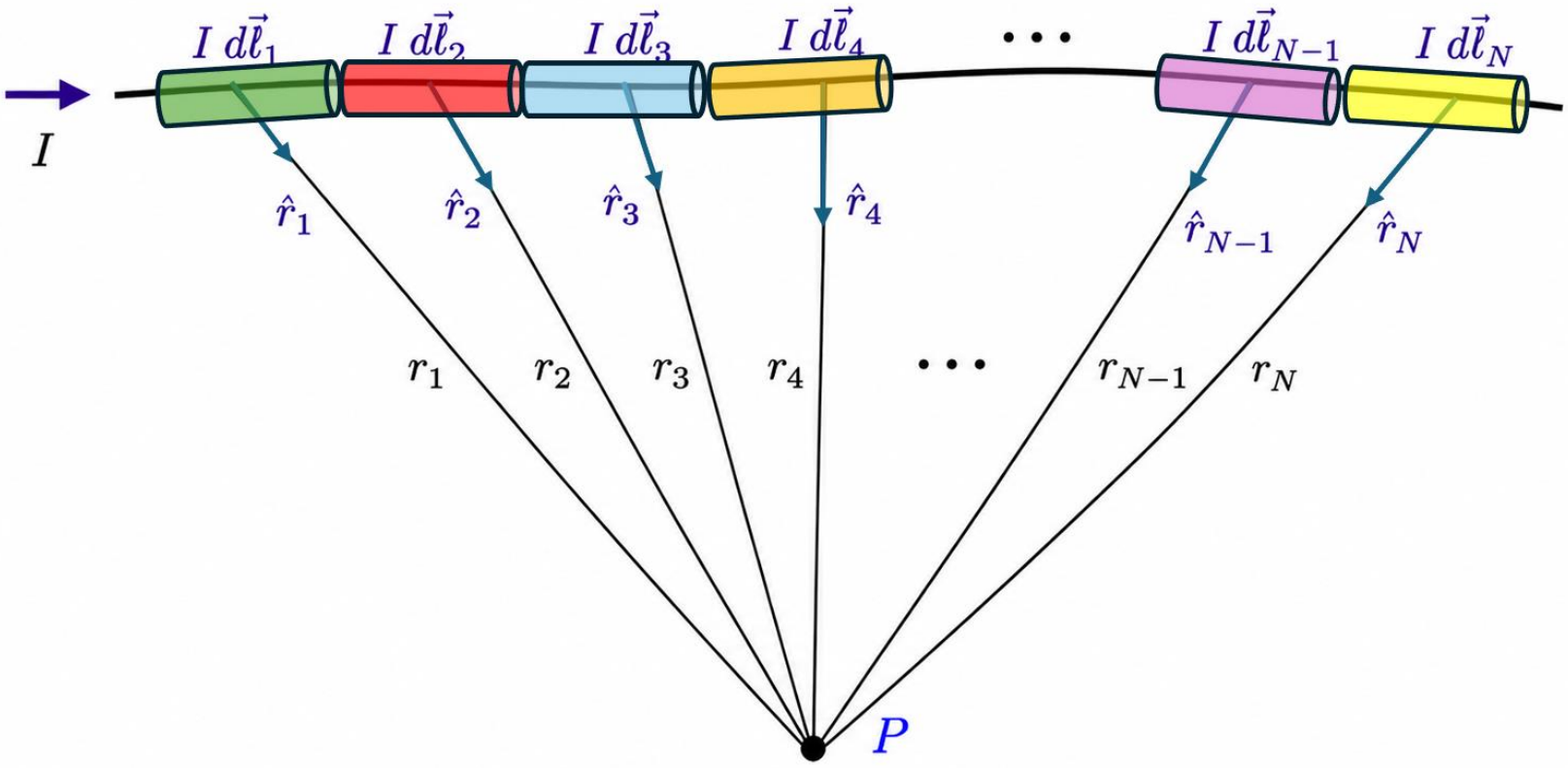
$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{dq\, \hat{r}}{r^2}$$



dq	=	charge element (differential charge)
$\hat{r}$	=	unit vector from charge element to field point P
r	=	distance from charge element to field point P
ϵ_0	=	permittivity of free space (8.854×10^{-12} F/m)

BIOT–SAVART LAW

A REAL WIRE IS MADE OF MANY SMALL CURRENT ELEMENTS



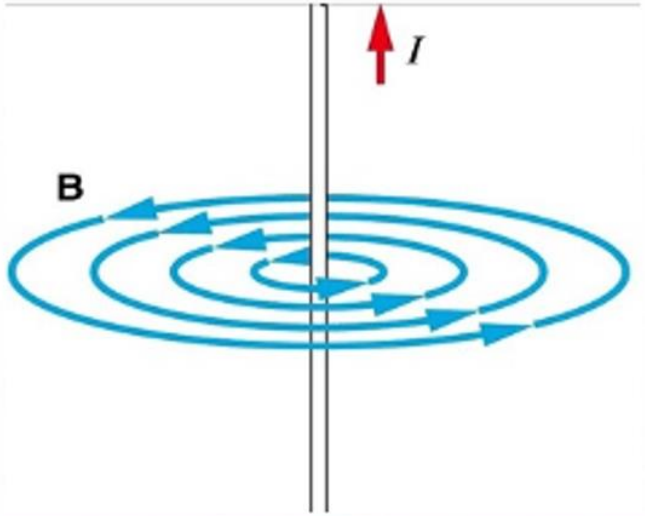
$$d\vec{B}_1 + d\vec{B}_2 + d\vec{B}_3 + \cdots$$

A tiny current element $I\,d\vec{l}$ produces a tiny magnetic field $d\vec{B}$.
The total magnetic field at point P due to the entire current distribution is found by integrating all contributions:

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{I\,d\vec{l} \times \hat{r}}{r^2}$$

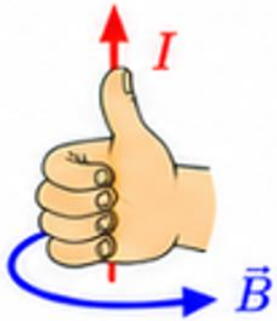
Magnetic field of straight wire and loop wire

1. STRAIGHT WIRE

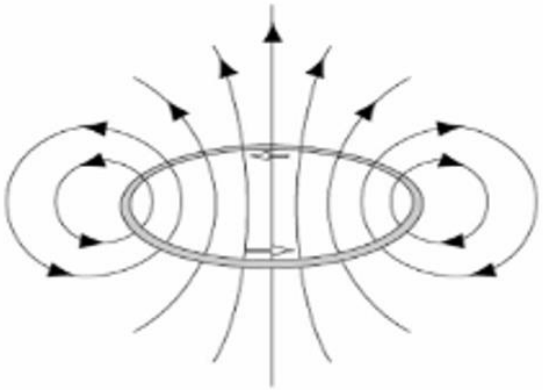


- Magnetic field lines are concentric circles around the wire.
- Direction given by the right–hand rule:

  Thumb along I (up), fingers give direction of $\vec{B}$.



2. CIRCULAR WIRE (LOOP)

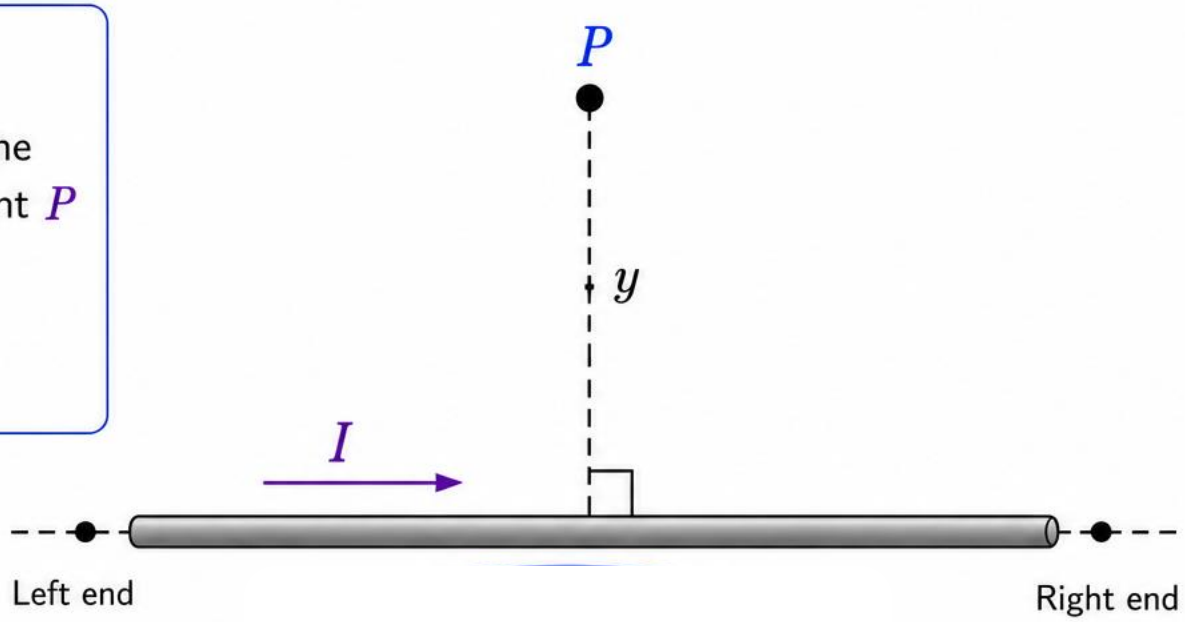


- Magnetic field lines pass through the center of the loop perpendicular to its plane.
- Outside the loop, they return around and form closed curves.
- Direction at the center given by the right–hand rule.

Magnetic Field of a Finite Straight Current-Carrying Wire

Question

How do we calculate the magnetic field at a point P produced by a finite straight wire carrying current I?



We will use the **Biot–Savart Law** to derive the magnetic field for this geometry.

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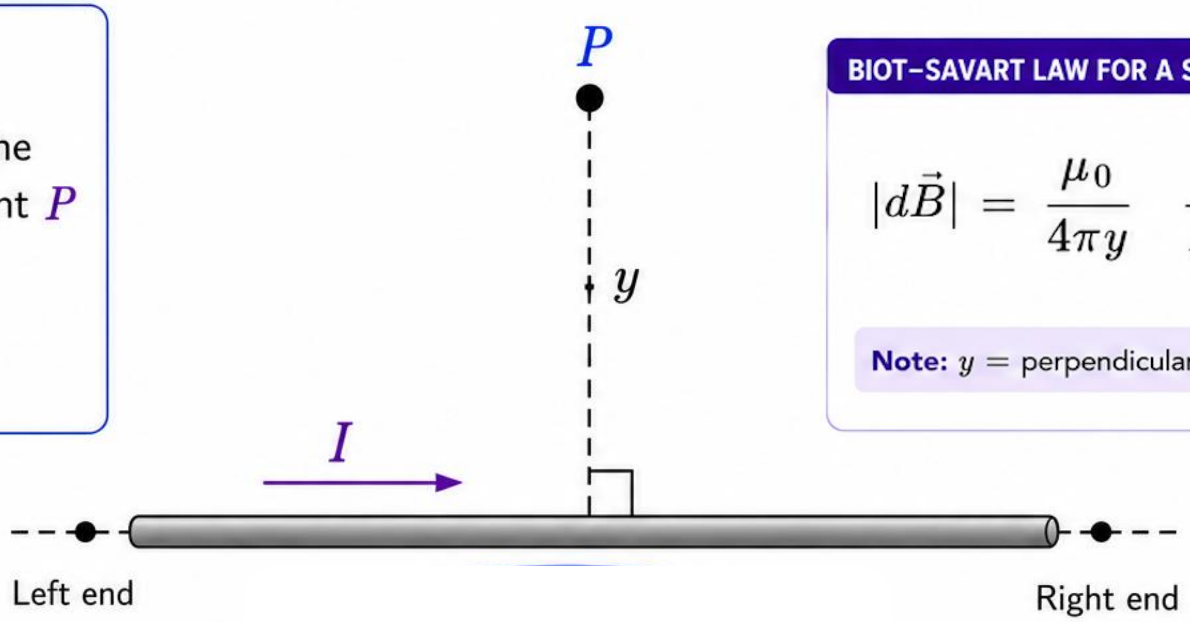


BIOT–SAVART LAW FOR A STRAIGHT CURRENT CARRYING WIRE

$$|d\vec{B}| = \frac{\mu_0}{4\pi y} \left. \frac{x}{\sqrt{x^2+y^2}} \right|_{\text{start point}}^{\text{end point}}$$

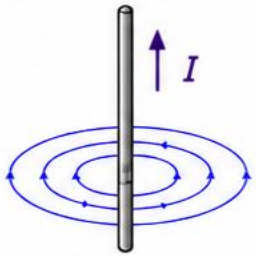
Note: y = perpendicular distance of the point to the wire.

We will use the **Biot–Savart Law** to derive the magnetic field for this geometry.

Magnetic field of a straight current-carrying wire and its extension to other geometric configurations.

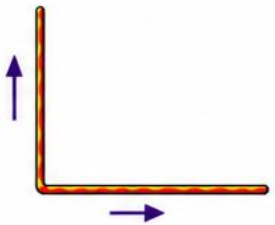
OPEN (NON-CLOSED) PIECEWISE-LINEAR WIRES

STRAIGHT WIRE

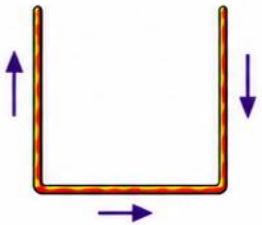


Field lines are concentric circles around the wire.

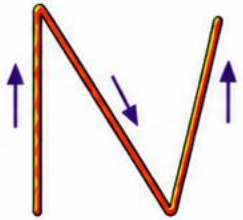
L-SHAPED WIRE



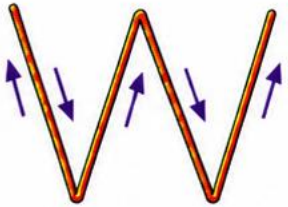
U-SHAPED WIRE



N-SHAPED WIRE



W-SHAPED WIRE



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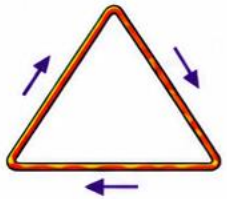
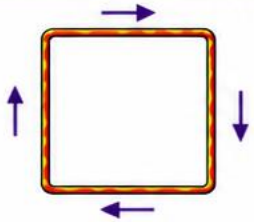
CLOSED CURRENT LOOPS (POLYGONS)

TRIANGLE LOOP



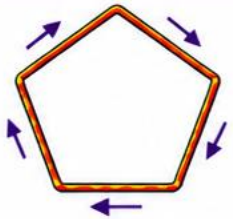
$N = 3$

RECTANGLE LOOP



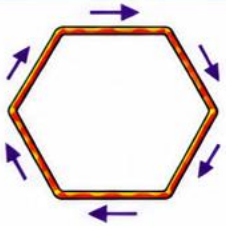
$N = 4$

PENTAGON LOOP



$N = 5$

HEXAGON LOOP



$N = 6$

...

These examples show that any current configuration made from straight wire segments—whether closed or open—can be analyzed using the magnetic field result for a single straight current-carrying wire.

N is the number of straight wires used to make up a certain type of loop.
(For example: $N = 3$ for triangle, $N = 4$ for rectangle, ...)

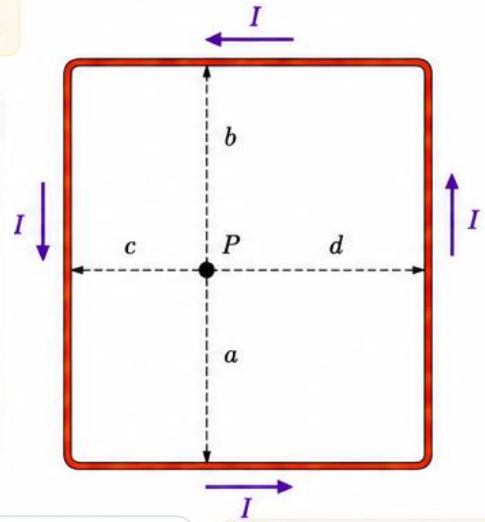
MAGNETIC FIELD AT POINT P DUE TO A RECTANGULAR CURRENT LOOP

Find the magnetic field at point P (inside the plane of the rectangular loop).

BIOT–SAVART LAW FOR A STRAIGHT CURRENT CARRYING WIRE

$$|d\vec{B}| = \frac{\mu_0}{4\pi y} \frac{x}{\sqrt{x^2+y^2}} \Bigg|_{\text{start point}}^{\text{end point}}$$

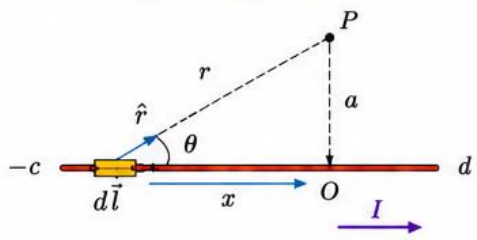
Note: y = perpendicular distance of the point to the wire.



APPROACH

1. Break the loop into four straight segments: bottom, left, top, right.
2. For each segment, place the origin at the point where the perpendicular distance from point P intersects the wire, and determine the beginning and end points of the segment relative to this origin.
3. Determine the direction of $\vec{B}$ using the right-hand rule, and apply the Biot–Savart law result for each segment.
4. Find the total magnetic field at point P by taking the vector sum of the magnetic field contributions from each segment.

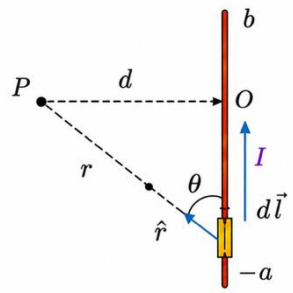
1) BOTTOM SIDE



$$\vec{B}_{\text{bottom}} = \frac{\mu_0 I}{4\pi a} \frac{x}{\sqrt{x^2+a^2}} \Bigg|_{-c}^{d} \hat{z}$$

Direction: out of page ($+\hat{z}$)

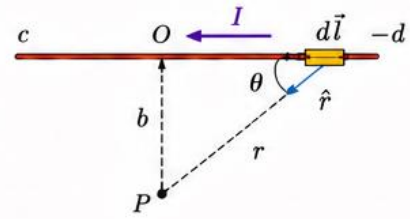
2) RIGHT SIDE



$$\vec{B}_{\text{right}} = \frac{\mu_0 I}{4\pi d} \frac{x}{\sqrt{x^2+d^2}} \Bigg|_{-a}^{b} \hat{z}$$

Direction: out of page ($+\hat{z}$)

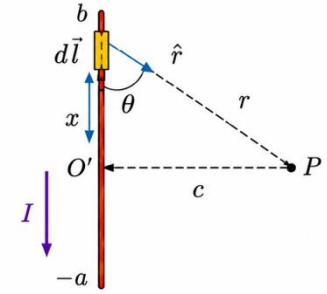
3) TOP SIDE



$$\vec{B}_{\text{top}} = \frac{\mu_0 I}{4\pi b} \frac{x}{\sqrt{x^2+b^2}} \Bigg|_{-d}^{c} \hat{z}$$

Direction: out of page ($+\hat{z}$)

4) LEFT SIDE



$$\vec{B}_{\text{left}} = \frac{\mu_0 I}{4\pi c} \frac{x}{\sqrt{x^2+c^2}} \Bigg|_{-b}^{a} \hat{z}$$

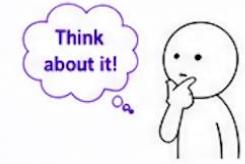
Direction: out of page ($+\hat{z}$)

TOTAL MAGNETIC FIELD AT POINT P

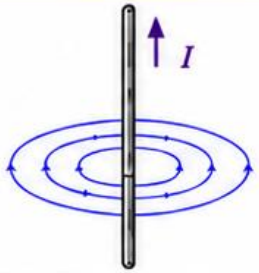
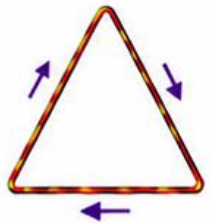
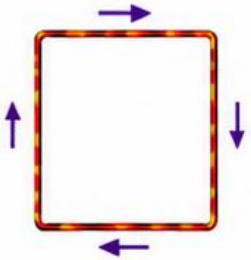
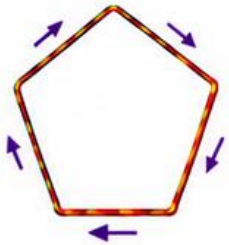
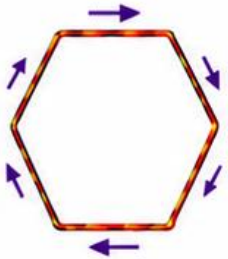
$$\vec{B}_{\text{total}} = \vec{B}_{\text{bottom}} + \vec{B}_{\text{left}} + \vec{B}_{\text{top}} + \vec{B}_{\text{right}} \quad (\text{along } \hat{z})$$

CHECK YOUR UNDERSTANDING

- ☐ What if point P was at the center of a square. How would this result simplify?
- ☐ How would the field change if P is very close to one side?

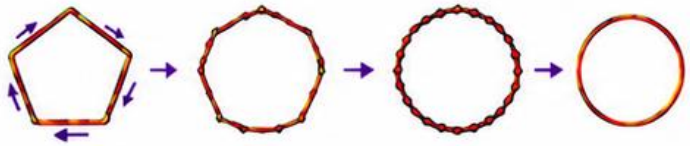


Magnetic field of a straight current-carrying wire and its extension to other geometric configurations.

STRAIGHT WIRE	TRIANGLE LOOP	RECTANGLE LOOP	PENTAGON LOOP	HEXAGON LOOP	...	CIRCULAR LOOP (LIMIT)
 Field lines are concentric circles around the wire.	 $N = 3$	 $N = 4$	 $N = 5$	 $N = 6$	• • • ...	$N = \infty$

1 From polygons to a circle

Break the straight-wire result into N pieces.
Let $N \longrightarrow \infty$ and each piece becomes tiny.

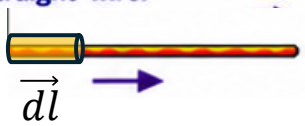


In the limit, we return to the differential (original) Biot–Savart law:

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{\ell} \times \hat{r}}{r^2}$$

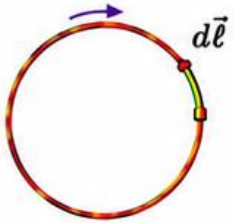
2 What changes?

Straight wire:



Integrate along a straight line.

Circular loop:



Integrate around an arc (angle ϕ from 0 to 2π).

We integrate over an angle, not a straight distance!

3 How hard is it?

Calculating the magnetic field at a random point inside or outside the loop:

The variables are no longer related in a simple symmetric way.

Difficult

Calculating the magnetic field at the center of the loop:

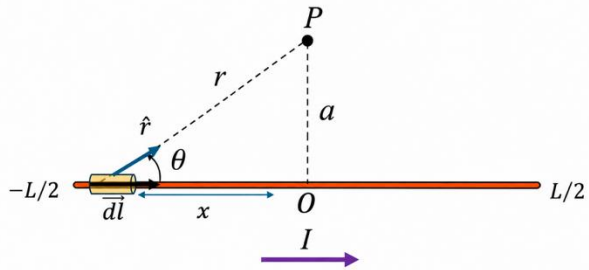
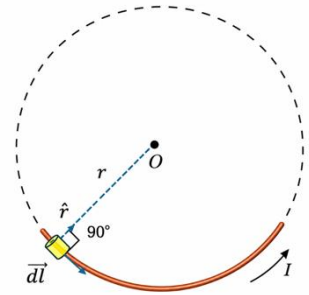
The variables are related in the simplest possible symmetric way.

Simple!

Key idea: Straight wire → sum along a **line**. Circle → integral around an **arc**. At the **center**, symmetry makes everything easy!

Magnetic field of a circular loop by comparison with a straight wire

Same idea, same steps — see where they differ!

	1. Straight wire (finite)	2. Circular loop	
			
1 Magnetic field due to an element	$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \hat{r}}{r^2}$	$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \hat{r}}{r^2}$	We always start by writing the small magnetic field due to a tiny piece of wire.
2 Setting up the problem	r is not given in the question, we need a coordinate system to define it.	r is the radius of the circle— it is given in the question.	✓ We do not need a coordinate system to define r.
3 Evaluate $d\vec{l} \times \hat{r}$	$d\vec{l} \times \hat{r} = dx \sin\theta \ \hat{z}$ ($\sin\theta$ varies along the wire)	$d\vec{l} \times \hat{r} = dl \sin 90^\circ \ \hat{z} = dl \ \hat{z}$ ($\sin 90^\circ = 1 \rightarrow$ constant)	★ Always perpendicular, so the direction and magnitude simplify!
4 Integrate to get $\vec{B}$	$\vec{B} = \frac{\mu_0 I}{4\pi} \hat{z} \int \frac{dx \sin\theta}{r^2}$	$\vec{B} = \frac{\mu_0 I}{4\pi r^2} \hat{z} \underbrace{\int dl}_{L}$ $\boxed{\vec{B} = \frac{\mu_0 I}{4\pi r^2} L \ \hat{z}}$ $\sin\theta$ and r^2 are no more variables	✓ Only dl is inside the integral.

Key takeaway:

Use the **same** Biot–Savart law and steps.

The circular loop simplifies because every element is the **same** distance from the center and always **perpendicular** to $\hat{r}$.

At the center

$$\boxed{\vec{B} = \frac{\mu_0 I}{2R} \hat{z}}$$

Direction: $\hat{z}$ (out of page) by right–hand rule

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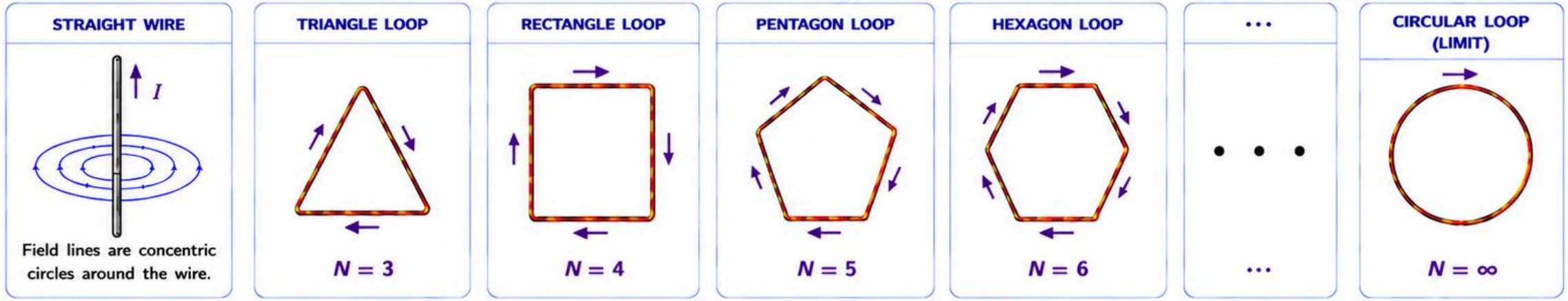

Magnetic field at y distance away from a current carrying straight wire

$$|d\vec{B}| = \frac{\mu_0}{4\pi y} \left. \frac{x}{\sqrt{x^2+y^2}} \right|_{\text{start point}}^{\text{end point}}$$

Magnetic field at the center of current carrying circular wire

$$\vec{B} = \frac{\mu_0 I}{2R}\,\hat{z}$$